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Research Proposal for the 2026 U.S. CMS HL-LHC S&C Program

Machine learning based particle flow reconstruction for the CMS Phase 2 detector

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1 Introduction

The particle flow (PF) algorithm [1] plays a central role in CMS event reconstruction. The present PF reconstruction uses information from the tracker, ECAL, HCAL, and muon spectrometer, iteratively links tracks and calorimeter clusters using proximity and other rule-based criteria, and builds a comprehensive list of final-state particles, called **PF candidates**. This list of PF candidates provides a global event description, which yields excellent performance for measuring jets, missing E_{T}, hadronic τ decays, electrons, photons, and muons.

PF reconstruction in CMS was established during Run 1, and it has been central to CMS analyses since then. For the upcoming HL-LHC operation, it needs significant updates. PF reconstruction for the Phase 2 detector needs to integrate high granularity calorimeter (HGCAL) reconstruction in the endcap region, incorporate forward tracks up to $|\eta| \sim 4$, and perform 4-dimensional reconstruction by incorporating the precision timing provided by e.g. the MIP timing detector (MTD) and HGCAL. It also should be able to cope with the increased data rates by exploring advances in modern computer hardware—particularly graphics processing units (GPUs).

The PI has spent the past few years on various aspects of PF developments for Run 3 and Phase 2 together with his collaborators. These developments include the PF rechit and cluster producer acceleration using GPUs [2, 3], the development of PF for the forward region ($3 \lesssim |\eta| \lesssim 4$) for Phase 2, and developments of "The Iterative CLustering" (TICL) for Phase 2 for the HGCAL region [4]. The TICL is a modular framework developed for heterogeneous infrastructure that provides the particle shower reconstruction and particle flow candidate reconstruction. It was primarily developed for HGCAL, but it could work well for other calorimeter regions, and thus can provide the coherent reconstruction across all calorimeter regions eventually. TICL has been used as the default PF reconstruction software for the HGCAL region since the time of Phase 2 upgrade HLT TDR [5], and it has been improved over time [6, 7].

I am grateful for the support from the 2023 U.S. CMS HL-LHC S&C program. With the support from this program, **Dr. Mohamed Darwish** joined Baylor from University of Antwerp in March 2024 to work on the project **"Developing heterogeneous particle flow reconstruction for the CMS Phase 2 detector"**, and made great progress in improving the TICL reconstruction. I would like to request support for him from the 2026 U.S. CMS HL-LHC S&C program as well in order to enable him to work on a new project **"machine learning based particle flow reconstruction for the Phase 2 detector"**, which is a natural extension of his work so far.

2 Recent progress related to particle flow reconstruction

In this section, we highlight a few developments made on PF reconstruction, which are relevant to the proposed project.

2.1 Porting the early steps of PF to GPU using a performance portability library

In the present PF reconstruction chain, one of the most time consuming steps is **PFClusterProducer**, which groups calorimeter hits to form clusters corresponding to showers created by incident particles. As the CMS HLT computing farm utilizes computing nodes equipped with GPUs, and more heterogeneous computing resources with GPUs are becoming available worldwide, we had opportunities to incorporate them into our reconstruction software, including PF. In event reconstruction including PF, many tasks can be expressed using "parallel" algorithms, and GPUs provide an ideal platform for parallel algorithms. Clustering is often not trivial to parallelize, but we adapted an excellent work by J. Jaiganesh and M. Burtscher [8] for PF clus-

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tering. The initial work by Dr. M. Saunders, who was supported by this program, was performed using NVIDIA CUDA; however, following the CMS wide transition to a portability library solution **Alpaka**, we re-implemented it in Alpaka, and commissioned it in time for CMS HLT operation starting in 2024, providing roughly a factor of 2–3 speed-up for PF clustering [2, 3].

The reconstruction of hadronic clusters from HCAL hits consists of forming clusters in each depth layer and then forming multi-depth clusters, linking layer-wise clusters to capture clusters from individual hadrons. The multi-depth clustering is another time-consuming step in PF, and the Alpaka version of it has been developed by a Fermilab scientist O. Strelchenko with support from M. Kortelainen and the PI's group. We expect that we will commission it in 2026 data taking, allowing us to gain more experience before we potentially use it for the barrel region during HL-LHC operation.

2.2 TICL development for the HGCAL region

For the Phase 2 PF reconstruction, we currently use a hybrid approach in which TICL is used for the HGCAL region and the traditional CMS PF algorithm in the other regions. The PI's group has been a part of this TICL development team, consisting of colleagues from CERN, LLR, ETH Zurich, RWTH Aachen, CNRS, Universita & INFN, Milano-Bicocca, and others. The flow diagram of TICL is shown in Fig. 1. It clusters HGCAL local hits into 2-dimensional layer clusters and 3-dimensional clusters called **tracksters** using the **CLUE** and **CLUE3D** algorithms [4, 6, 7], links HGCAL tracksters to tracks from the CMS tracker, and provides PF candidates, as shown in Fig. 1. The current Linking step of TICL is performed with a fully rule-based algorithm. TICL is also used to provide inputs to the dedicated reconstruction chain for electrons and photons [5].

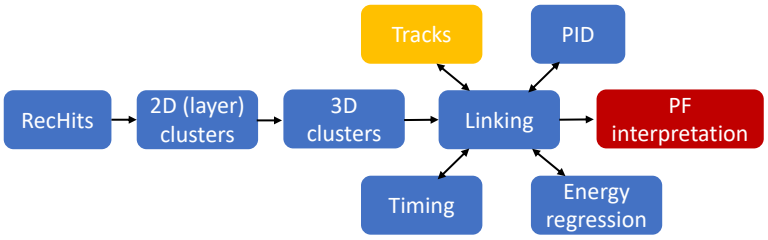

Figure 1: Illustration of the building blocks of the HGCAL TICL reconstruction workflow.

Since Darwish joined the Baylor group in March 2024, he has been actively involved in the TICL development team. After he became familiar with TICL, he worked on various aspects of TICL, including the development of a plugin for use of ML models in TICL and an ML model for classifying calorimeter clusters into electromagnetic or hadronic showers, and energy regression of calorimeter clusters based on a CNN. These developments are already fully integrated into CMS software. He is currently working on improving links between tracks and TICL calorimeter tracksters using a graph neural network model.

The TICL framework has been developed targeting the heterogeneous computing architecture, with data structures optimized for GPUs, and good progress has been made already on that front. The Alpaka version of CLUE is already in CMSSW as an external generic library. An evolution of this library is also in the process of being integrated into CMSSW to take care of CLUE3D for 3D clustering. However, no progress is made on running the Linking step on heterogeneous platforms.

2.3 MLPF: Machine learning based particle flow reconstruction

There has been a very exciting and promising development on an ML-based PF (MLPF) reconstruction algorithm [9, 10]. Its development and integration into CMSSW has been performed by the MLPF team, consisting of people from NICPB, CERN, UCSD, Purdue, Brown, and CUA, and led by J. Pata, J. Duarte, and F. Mokhtar *et al.* It has been extensively tested with CMS simulation data and collision data under the Run 3 data taking condition [11].

The most recent version of MLPF is based on a transformer-based architecture using self-attention for PF reconstruction, where the model predicts all reconstructed particles simultaneously in one inference step. The architecture is illustrated in Fig. 2. The model employs a plain global transformer, but it is accelerated with FLASHATTENTION so that there is no quadratic scaling. The inputs to the model are the reconstructed tracks and calorimeter clusters in an event, and the model outputs are reconstructed particle candidates.

The input and target sets generally have a different number of elements, so the authors define a per-particle loss function by first assigning each target particle to exactly one unique input element, referred to as its *primary element*. This assignment is

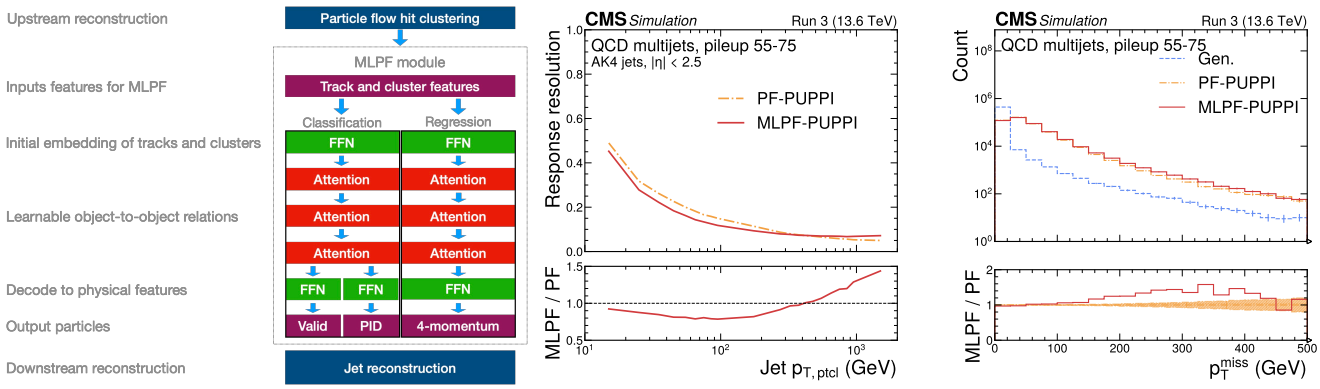

Figure 2: (left) The neural network structure of the MLPF model [11]. Input and output features are visualized in magenta. The last dimension contains a binary output to predict the presence of a particle (Valid), multi-class outputs for the PID, and the predicted momentum values. The jet resolution (middle) and uncalibrated $p_{\mathrm{T}}^{\mathrm{miss}}$ distribution (right) in QCD multijet validation samples with 55–75 pileup interactions.

made from the set of target particles to the set of input elements (tracks or clusters): charged particles are assigned to their originating tracks, and neutral particles to the calorimeter clusters where they deposit most of their energy. The per-particle loss function consists of three terms: a binary classification loss term, a multiclass particle identification (PID) classification loss term, and a regression loss term for the particle four-momentum. The binary classification term quantifies whether the predicted particle's corresponding track or cluster is the primary element for a given target particle.

The model is trained on CMS Run 3 full simulation data sets that include realistic pileup conditions, and its performance is evaluated in $\mathrm{t\bar{t}}$ and QCD multijet testing samples, as well as on the actual collision data, with promising results. As for jet-level performance, MLPF seems to achieve comparable or improved performance compared the standard PF in general, as shown in Fig. 2. In the high-p_{T} region, the relative improvement of MLPF decreases, which may be attributed to the limited number of high-p_{T} training samples. The $p_{\mathrm{T}}^{\mathrm{miss}}$ performance is found to be generally consistent between PF and MLPF, with MLPF reconstructing a somewhat harder $p_{\mathrm{T}}^{\mathrm{miss}}$ spectrum in the tails.

3 Machine learning PF reconstruction for the Phase 2 detector

As discussed above, good progress has been made in recent years for calorimeter hit clustering for CMS HCAL and HGCAL as well as machine learning PF reconstruction (MLPF) with tracks and calorimeter clusters as inputs. MLPF has been studied under the Run 3 data taking condition so far. The proposed work is to **adapt MLPF-like approach for the Phase 2 detector scenario under the HL-LHC high pileup condition**.

For the Phase 2 PF reconstruction, we currently use a hybrid approach in which TICL is used for the HGCAL region and the traditional PF algorithm in the barrel and forward regions.

The PI has started discussion with the rest of the TICL group and MLPF group about the process of integrating MLPF to the Phase 2 scenario. **The general consensus for the initial step is to adapt the current Run 3 MLPF for Phase 2.** For the HGCAL acceptance region, we will use the TICL tracksters as inputs to MLPF. This can be achieved by using MLPF for the Linking step of TICL (see Fig. 1). As for the barrel and forward regions, we can start with the current PF clusters as inputs to MLPF, which means we run MLPF in separate paths, inside and outside the HGCAL regions, and we merge the PF candidate outputs at the end. There is an ongoing effort to extend the TICL's CLUE and CLUE3D to the barrel region within the TICL group. When this effort is at the mature stage, we will test using them as inputs to MLPF as well.

After we define the baseline Phase 2 MLPF in this way, we will perform detailed performance study, and we also plan to explore further improvement ideas. The private communication with the MLPF team suggests that the extrapolation of the current MLPF timing will lead to around 100 ms of processing time and 600 MB of GPU RAM for inference for an event

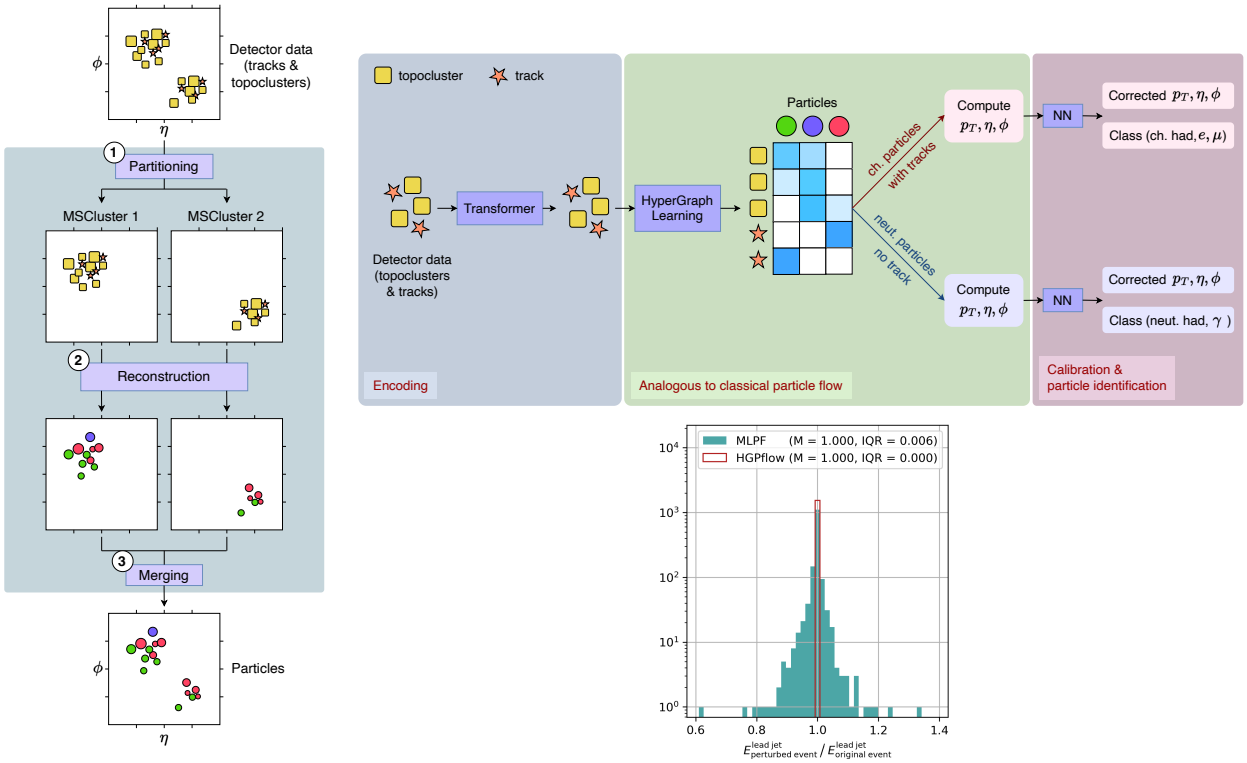

Figure 3: (left) The event partitioning procedure using Mean Shift clustering in the η-ϕ plane. (upper right) Schematic showing the three main blocks of the HGPflow algorithm. (lower right) Test of the locality of the model predictions in dijet events simulated in CLIC. The ratio quantifies the dependence of leading jet energy to subleading jet energy when a perturbation of -80% is applied to the latter.

with 200 pileup interactions on a node with NVIDIA L4 GPU. It will not be trivial to use much more complex ML models while keeping the timing and memory in a reasonable range, but one proposed idea is to use *e.g.* neural architecture search to find a better configuration within the same budget.

There have been another interesting development called **Hypergraph Particle Flow (HGPflow)**, a particle reconstruction algorithm based on learning hypergraphs [12]. This is a separate ML approach for PF reconstruction, and differences from MLPF include: (1) partitioning of an event in order to localize PF reconstruction task using a *mean shift* clustering algorithm, and (2) using a learnable matrix relating the input and output sets, without defining a primary input element for a given target particle. These points are illustrated in Fig. 3.

The event partitioning has a possibility to make the model computationally efficient and address the **locality** issue, *i.e.* avoids potential bias from long-range physical correlations, with added complexities of partition edge effects. For particle reconstruction to function universally, it would be best to approach it as a local problem, agnostic to long-range correlations. If reconstructed PF candidates provide information that is biased toward the physics process used for training, they no longer serve their purpose as generic inputs for event reconstruction. Ref. [12] provided a quantitative way to measure locality, and the version of MLPF they tested showed a small but definite long-range correlation (see Fig. 3). We plan to perform this locality test in the most up-to-date of MLPF. We could potentially address the model locality by trying HGPflow, but also in MLPF by partitioning the event before the correlation-building layers. The exact path forward will be decided after more discussions with other collaborators.

The event partitioning allowed HGPflow to employ a more complex model than the current MLPF. HGPflow formulates the PF reconstruction as a hypergraph learning task [12]. If we can manage the partition edge effects for particles around the partition boundaries, the HGPflow model may outperform MLPF physics-wise. We will still need to carefully evaluate the computational scalability up to the HL-LHC high pileup condition, and we plan to look into it as a part of this proposed

project.

4 About the postdoctoral candidate

Dr. Mohamed Darwish joined Baylor at the beginning of March 2024, and he ramped up quickly in the TICL group. He had a strong interest in utilizing ML techniques in TICL. After he became familiar with TICL, he worked on various aspects of TICL, including the development of a generic plugin for ML use in TICL and an ML model for classifying calorimeter clusters into electromagnetic or hadronic showers, and energy regression of calorimeter clusters based on a CNN. His familiarity with TICL and ML will make a good candidate to work on this proposed task. He currently serves as a CMS PF group L3 convener for reconstruction, and Offline and Computing (O&C) group L2 convener for Phase-2 upgrade software.

5 Timeline and milestones

The tentative timeline and the milestones of the planned work are presented below. Please note that the timeline and milestones will need to be adjusted based on outcomes of the earlier steps and various activities in the CMS Collaboration as a whole.

March 2026: the first implementation of MLPF for the Phase 2 detector with TICL tracksters for HGCAL and current PF clusters for other calorimeters as inputs, prepare the training and validation data sets

June 2026: MLPF training, physics and computational performance studies including locality tests

September 2026: neural architecture search, optimization of the model, scale-up of training data sets

December 2026: the first implementation of HGPflow for the Phase 2 detector, prepare the training and validation data sets

March 2027: HGPflow training, extensive comparisons of traditional PF, MLPF, and HGPflow

6 Summary

In summary, the particle flow algorithm plays a central role in CMS event reconstruction, and its upgrade for Phase 2 is of paramount importance for the CMS experiment's success during HL-LHC. I think the proposed work on the development of the machine learning based particle flow reconstruction for Phase 2 has significant impacts on the U.S. CMS HL-LHC S&C program and the CMS Collaboration as a whole. Thank you for your consideration.

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